COMPUTER SYSTEM DESIGN
COURSE: CSL559
ASSIGNMENT 01: PHASE 2 REPORT

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GITHUB URL:
https://github.com/SandstromPL/AI-based-Village-Pond-Planning-System-.git

WORKING API ROUTE URL:
- URL head: http://10.1.75.53:3278
- POST: http://10.1.75.53:3278/api/v1/analyzeContour
- Health check: http://10.1.75.53:3278/api/v1/health

Catchment Estimation Approach

The system estimates catchment area from the uploaded contour KML/KMZ file using a terrain-and-hydrology pipeline:

1. Contour LineStrings and their elevation values are extracted from the KML.
2. Geographic coordinates are projected to UTM so that distances and areas are calculated in metres.
3. The contour data is interpolated into a 30 m resolution Digital Elevation Model (DEM).
4. Artificial terrain pits are removed using the Priority-Flood depression-filling method.
5. D8 flow direction is calculated: every DEM cell drains to its steepest lower neighbouring cell.
6. Flow accumulation identifies cells where water from many upstream cells converges.
7. For each pond candidate, reverse Breadth-First Search is performed from that candidate cell through the flow-direction graph. All cells that eventually drain into it form its catchment.

8. Catchment area is calculated as:

$$\text{Catchment area} = \text{upstream cell count} \times \text{DEM cell area}$$

The catchment boundary is converted into a GeoJSON polygon for map visualization. This approach avoids the incorrect assumption that the lowest point is automatically the best pond location.

Demonstration Using the Provided Contour Map

The provided contours_1m.kml file was uploaded through the API and analysed successfully.
Analysis Summary Table:

- Contours extracted: 1,355
- Input elevation range: 267 m to 298 m
- DEM resolution: 30 m
- DEM grid size: 106 × 130 cells
- Average terrain slope: 1.86°
- Pond candidates returned: 5
- Recommended candidate: C1
- Recommended location: 21.263169° N, 81.282811° E
- Candidate score: 82.8 / 100
- Slope at candidate: 1.97°
- Catchment area: 0.045 km²
- Flow accumulation: 49 cells

Natural depression depth: 1.33 m

Candidate C1 was selected because it had the largest catchment and flow-accumulation values among the evaluated candidates, a gentle slope suitable for construction, low elevation relative to alternatives, and a measurable natural depression.

The API response includes:

- Parsed input metadata
- Terrain statistics
- Ranked pond candidates
- Recommended pond location
- Catchment boundaries
- Contour lines, candidates, and catchments as GeoJSON layers
- Analysis assumptions and thresholds

Rainfall, runoff estimation, and pond-size calculations are currently marked as Phase 3 placeholders. Therefore, the present result is a terrain-and-catchment suitability recommendation.

API Documentation

The backend provides automatic interactive API documentation through FastAPI Swagger UI:
http://<HOST>:<PORT>/docs

Live URL - http://10.1.75.53:3278/docs

Available endpoints:

Method	Endpoint	Purpose
POST	/api/v1/analyzeContour	Upload a KML/KMZ contour file and perform full terrain/catchment analysis.
GET	/api/v1/analysis/{id}	Retrieve a previously generated analysis result.
GET	/api/v1/health	Check backend service availability.

For the required analysis endpoint, the uploaded multipart file variable is exactly:
contour_map

Calling the API Using Postman

1. Open Postman and create a new HTTP request.
2. Set the request method to POST.
3. Enter the API URL:
   http://10.1.75.53:3278/api/v1/analyzeContour
4. Open the **Body** tab.
5. Select **form-data**.
6. Add one form field:
   - Key: contour_map
   - Type: File

        - Value: Select the contour .kml or .kmz file from the local machine
    7. Optionally add the following header:
        - Accept: application/json
    8. Click **Send**.

The server receives the uploaded contour file under the required variable name contour_map, validates and parses it, performs terrain and catchment analysis, generates candidate pond locations, and returns a formatted JSON response.

The response includes input metadata, terrain statistics, ranked candidates, the recommended location, catchment boundaries, GeoJSON layers for map visualization, and analysis assumptions.

AI usage:
AI tools were used responsibly during the project.
ChatGPT was used to understand the assignment requirements and prepare the initial design, architecture, and algorithm-planning documents.
Antigravity was used to assist with the initial backend implementation.
Codex was used for debugging, output formatting, minor final enhancements, and report preparation. All generated work was reviewed, tested, modified where necessary, and understood by the project author(me).

**** This report was prepared with assistance from AI tools ****